



RESEARCH & INSIGHTS HUB

Monthly Outlook: Machine Learning Models

Has crypto market performance in 1H24 been driven more by macro factors or fundamentals?

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Linear regression and random forest models are two commonly used machine learning algorithms that help provide valuable insights into the factors influencing cryptocurrency price movements. The difference is that where linear regression models provide a relatively *straightforward* interpretation of how certain factors directly impact cryptocurrency values, random forest models capture the intricate, *non-linear* relationships and interactions between various input factors. Ultimately, our goal is to determine whether the relationships that may have driven crypto performance in the first half of 2024 could be relevant in the second half of this year.

Our results suggest that over the last six months, the core drivers of BTC, ETH and SOL performance have been predominantly unique to those tokens, such as network fees and total value locked – or spot ETF flows in the case of bitcoin. That is, despite higher correlations between crypto and traditional asset classes in recent months, the most salient variables influencing crypto returns still tend to be more idiosyncratic in nature. This suggests that cryptocurrencies offer important portfolio diversification benefits, such as mitigating risk and enhancing overall returns, per our previous work on this topic.

However, there are some things that we weren't able to fully capture in our models, such as shifts in the US regulatory environment. For example, we think the House of Representatives' decision to approve the Financial Innovation and Technology for the 21st Century Act (FIT21) may have contributed to a change in market sentiment earlier

this year. With the US elections likely to be an important driver in 2H24, this could be a notable omission.

For more details, see our [full report](#).

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